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ORIGINAL RESEARCH

Diminishing or increasing? Exploring the impact of online movie reviews on consumer demand within and across two sequential release stages

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Abstract

This study examines the impact of online movie reviews on consumer demand within and across two sequential release stages: theatrical run and online streaming. In contrast to extant findings of a diminishing sales effect of online reviews, we observe an increasingly positive impact of movie reviews in the cinema release stage, in line with the theory of diffusion of innovations. Moreover, consumers are predominantly conservative in using product reviews to select foreign products due to the cultural gap. After movies are released online, the role of product reviews changes according to the release gap. Movies with a higher review valence and volume during the cinema release will continue to benefit from the reviews with a shorter release gap. In contrast, movies that receive a lower review valence in the early market benefit from setting a longer release gap and waiting for the online buzz to dissipate.

Keywords Online product reviews · Product life cycle · Motion pictures · Cultural product · Cross-stage release gap

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1 Introduction

Business researchers have extensively studied the sales impact of online product reviews–a type of word of mouth (WOM) (e.g., Chevalier & Mayzlin, 2006; Liu, 2006). Whereas offline WOM is usually perishable over time (Elberse & Eliashberg, 2003), online reviews are documented on websites. Thus, potential buyers can easily access them at any time (Chevalier & Mayzlin, 2006), meaning they can see previously accumulated product evaluations. In line with this logic, the sales impact of online WOM should be persistent, if not increasing, over time.

However, extant studies analyzing the temporal role of online reviews reveal a diminishing sales effect, such that their impact on consumer demand decreases over time (e.g., Cui et al., 2012; Hu et al., 2008; Liu, 2006). The authors attribute this result to consumers' reduced product interest in the late periods of the product life cycle (PLC) – proxied as the amount of time since release. Perhaps for this reason, many sellers focus on gathering positive reviews during the early period of a product release, even if the reviews may not be truthful (He et al., 2022). Yet, such a strategy may depreciate the value of online reviews, leading consumers to make less informed purchase decisions.

Do online product reviews indeed have a decreasing impact on sales? We take two perspectives to address this question. The theory of diffusion of innovations (Rogers, 1962) characterizes early consumer groups (i.e., innovators and early adopters) as less risk-averse and often opinion leaders in the sense that they share their evaluations to influence future buyers. By contrast, late consumer groups (i.e., the majorities and laggards) are more conservative and tend to rely heavily on early consumer groups' opinions. They may only adopt a product after it receives enough positive WOM. Accordingly, online consumer opinions – usually cumulated and saved on the review platforms – should have an increasing (rather than diminishing) effect on sales throughout the PLC.

From a data analytics perspective, we see two challenges regarding the temporal role of product reviews that extant studies have not addressed. First, observing the entire life cycle of many products is difficult. For example, a book can be on the market for decades; in this case, estimations using a sample with an arbitrarily truncated observation window may lead to biased results. Second, when data on the complete life cycles of products are available, researchers often get an imbalanced panel because products with poor sales performance usually quit the market early. Separating the data by time (e.g., week) and applying an ordinary least squares (OLS) model (e.g., Moon et al., 2010) may produce less significant estimates in the late periods due to the reduced variations in the review measures since most remaining products are popular on the market and thus have similar review features (e.g., the variance of average rating is limited). Taking these factors into account, the findings of a decreasing sales effect of reviews may be inconclusive.

To overcome the data analysis challenges, we employ several rigorous approaches to understand the relationship between product reviews and consumer demand over time. First, similar to Liu (2006), we conduct research in the context of the motion picture industry. Movies have a short cinema release window;

thus, we are able to observe the sales performance during their entire life cycle. Second, we apply the product-level fixed effects specification to the panel data to mitigate the concern that unobserved time-invariant factors (e.g., movie genre or pre-release factors, such as production budget and critic reviews) may bias the estimates. More importantly, we use a Heckman-style two-stage model (Heckman, 1979) to handle the incidental selection issue (i.e., products with poor sales performance drop from the panel early). Third, instead of splitting the data by time and comparing estimates across the subsamples, we run models using the complete panel and taking interactions between product age and the review features, including review valence (i.e., the average rating), volume (i.e., the total number of reviews), and the interaction between them. These steps help draw a complete picture of online product reviews' sales impact over time.

By connecting the box office revenue and viewership data with online WOM from a movie review platform in China, we examine the effect of product reviews on consumer demand throughout the movie's cinema release window. The results suggest an increasingly positive impact of product reviews. Specifically, the effect of review valence on consumer demand increases from the opening day to the late periods of the theatrical run. Moreover, movies with a higher review valence *and* volume (i.e., positive endorsements by a vast number of buyers) gain additional benefits from reviews in the late periods. The findings imply that the overemphasis on generating (positive) reviews during the early period cannot improve sales in the long run; *only* products that continue to receive positive reviews can do so.

We also compare the impact of reviews on consumer demand for domestically produced Chinese movies and imported American movies. American movies, especially those produced by Hollywood studios, are widely known to have a global impact (Scott, 2002). Our data show a huge fan base for Hollywood hits such as *The Hobbit* and *The Amazing Spider-Man*. As such, consumers may adopt less restrictive selection criteria for American movies due to their popularity and high product awareness. The estimated results indicate that this conjecture is the case on the opening day: neither review valence nor its interaction with review volume matters. However, late consumer groups are much more conservative when choosing American movies than when choosing their Chinese counterparts. The sheer increase in the average rating cannot improve sales. Instead, consumers only consider movies with high levels of *both* review valence and volume, and such an interaction effect increases over time. American movies are produced in a culture unfamiliar to many late adopters in China. They may not have enough cultural background and knowledge to appreciate such products unless they receive substantial positive comments, which indicate that the movies are acceptable to most audiences in their culture. Overall, these findings further support our proposed conjecture that the impact of online reviews on consumer demand increases over time.

What about the impact of product reviews in an even later stage–when the movies are released online[1]*?* On the one hand, although the product is released on a new

[1] We use the term "stage" to distinguish the two life cycles of movie release (i.e., *in theater* and *online*). By contrast, we use the term "period" to refer to the change of time within a product's life cycle.

platform, its sales growth can be regarded as a continuous process from the previous stage because it is exposed to new consumers in the second market.[2] If the consumers in this stage are even more risk-averse, we should observe a stronger impact of product reviews on their demand. On the other hand, compared with moviegoers in the cinema, online users have very different consumption experiences. They only need to subscribe to a video streaming platform to access all movies on the site, and a monthly payment is often cheaper than a single movie ticket. Also, they can stop watching a video at any time if they do not enjoy it. In short, watching an unfavorable movie online is less costly than at the cinema. As such, consumers may lower their product selection criteria and be more tolerant of negative opinions. If this is the case, the impact of reviews on consumer demand may become less significant over time, indicating a qualitative difference between the two stages.

Our estimation results of the online stage reveal two distinct scenarios: (1) When movies are released online shortly (e.g., within two weeks) after the theatrical window, review valence and its interaction with review volume continue to have an increasingly positive impact on viewership. (2) However, when the gap is substantially large, review valence no longer has a direct effect; instead, consumers seem to choose movies with a higher review volume.

We provide several explanations for these findings. When the release gap is short, review platform users who plan to watch a movie online during its cinema release stage will be the next group of consumers. These individuals are essentially the "later" adopters in the previous market, and their impressions of the movie are still associated with its online buzz during the cinema release (Luan & Sudhir, 2022). As a result, their adoption decisions are strongly affected by WOM from the early stage. By contrast, when the release gap is large, those on the "waiting list" will likely forget about the product due to its unavailability for a significant period. In this scenario, the second-stage moviegoers will often be a new consumer group who has not followed reviews during the theatrical run. Moreover, video streaming platforms often recommend movies that have received high awareness, as indicated by the volume of reviews. As such, the estimate of review volume in the online stage implies a correlation rather than an actual impact of review volume on consumers' choices. We conduct descriptive analytics to support these conjectures.

The findings in this study offer insightful business implications. First and foremost, the sales impact of online product reviews does not decrease but increases over time. Only products with a high review valence (and, in some cases, a high valence *and* volume) would benefit from the online WOM in the long run. Thus, we discourage sellers from soliciting positive reviews during the early period of product release. Such a strategy may help generate short-term demand, but it can eventually hurt sales performance and even shorten the PLC (if the product does not receive positive reviews consistently). It is more effective to focus on producing high-quality products that can consistently receive positive WOM. Second, the market is more

[2] Our study uses user-reported viewership data for the online stage estimations. Because users cannot report their repeat watching behavior on the platform, we focus on analyzing the demand of first-time buyers.

competitive for foreign products due to consumers' unfamiliarity with the culture. These products may only succeed if they receive sizable positive reviews over time. Third, for products that can be released on a different platform, sellers of products with low review valence in the early market should consider a substantial release gap. As such, they may gain more sales in the late stage because the new consumer groups might be qualitatively different from those in the early stage and may not follow their opinions.

Overall, we contribute to the literature on online WOM by unveiling the increasingly positive impact of product reviews on consumer demand. This unique finding is based on models that handle the incidental selection issue that extant studies have not addressed. Furthermore, we study how consumers utilize reviews when selecting products produced in a foreign culture (versus domestically produced products). Finally, our estimations using demand data across the two product release stages – in different markets and platforms – shed light on when online reviews have a qualitative change in their impact on sales. The related empirical findings provide actionable intelligence for marketers to optimize their campaigns during the critical windows of a movie's release.

The remainder of the paper is organized as follows. We present the related literature in Sect. 2, focusing on studies that address the sales impact of online reviews *over time*. Section 3 details the development of the hypotheses. The data and model are described in Sect. 4, followed by the main results in Sect. 5 and mechanism explorations in Sect. 6. We conclude the paper with the business implications and limitations of the research in Sect. 7.

2 Literature review

2.1 Temporal change in the sales impact of online product reviews

Research on WOM in the marketing literature is not scarce. Studies have covered a wide range of topics regarding the effect of WOM. For example, Punj (2013) investigates whether consumers who conduct online product research are also more likely to post online product reviews. In a study about online reviews for vacation rentals, Zervas et al. (2021) collect data on properties listed on Airbnb worldwide and find that nearly 95% of Airbnb properties boast an average star rating of either 4.5 or 5 stars (the maximum); virtually none have less than a 3.5-star rating compared with the ratings of roughly 700,000 hotels, B&Bs, and vacation rentals that were listed on TripAdvisor. In a recent study, Xu et al. (2023) predict consumers' likelihood of creating and seeking online WOM.

In contexts that are more similar to ours, earlier studies focused on the effect of online WOM communications on actual sales in many markets, such as books (Chevalier & Mayzlin, 2006), electronics (Cui et al., 2012), and movies (Duan et al., 2008). In this section, we review the set of research most relevant to the topic of our study – the impact of online product reviews on consumer demand *over time* (summarized in Table 1).

Table 1 A summary of extant studies of the impact of online product reviews on consumer demand over time

Authors	Product review data source	Product review measures	Key findings	Cross-stage comparison	Cultural production
Godes and Mayzlin (2004)	TV comments from Usenet[a]	Volume and dispersion of online conversations about TV shows	Dispersion of messages can increase the viewership of the next episode[b]. This impact drops for future episodes	No	No
Liu (2006)	Yahoo Movies message board[a]	Weekly volume of posts and percentage of positive and negative messages	The weekly volume of posts has a positive effect on box office revenues. Such an effect decreases over time	No	No
Hu et al. (2008)	Reviews of books, DVDs, and videos on Amazon.com	Average rating, review quality (number of useful votes), and reviewer exposure (up-to-date posts created by the reviewer)	The difference in sales between favorable and unfavorable reviews is significant only for products in the early period of their life cycle	No	No
Moon et al. (2010)	Yahoo Movies message board[a]	Average rating	The average rating only affects the box office revenues in the opening week	No sales comparison	No
Cui et al. (2012)	Reviews of electronics and video games on Amazon.com	Average rating, number of ratings, and percentage of positive and negative reviews	Review volume has an inverted U-shaped relationship with product sales, suggesting a diminishing effect	No	No
Gopinath et al. (2014)	Online WOM of cellular phones from Howard Forums	Valence of attributed-oriented, recommendation-oriented, and emotion-oriented WOM	Only the valence of recommendation-oriented WOM can affect sales, and such an impact increases over time	No	No

Table 1 (continued)

Authors	Product review data source	Product review measures	Key findings	Cross-stage comparison	Cultural production
This paper	Movie reviews from Douban	Average rating, number of ratings, and their interaction	The effect of review valence and its interaction with review volume on product demand increases *over time* and *across stages* (with a short release gap)	Yes	Yes

[a]In these articles, consumers did not directly observe summary statistics of average rating and rating volume in the data source

[b]The viewership in this paper is a rating score, not the actual sales-related measures

Godes and Mayzlin (2004) measure the user posts on Usenet and analyze the impact of volume and dispersion of online conversations on the viewership of TV shows. They identify a positive effect of dispersion on the early episodes but also show that such an effect decreases for future episodes. This diminishing effect is logical, given that consumers watching later episodes should be those who like the show and have obtained enough product information from the early episodes. The unique feature of TV shows (i.e., they are episodic) differs from the product in our study in that movie audiences usually have no product experience prior to purchase.

Liu's (2006) and Moon et al.'s (2010) studies have the most similar research setting to ours. Both papers analyze the box office effect of reviews from Yahoo Movies. While Liu (2006) discovers a positive impact of review volume during the first six weeks of cinema release, Moon et al. (2010) observe a direct effect of review valence only in Week 1 of a movie's cinema release. Liu (2006) ascribes the diminishing effect to consumers' losing interest in the later periods. Interestingly, both studies split the data by week and applied OLS regression models to the cross-sectional observations. Such methods have a few possible limitations. First, the variation of review measures in the late periods may become limited since movies surviving in the market tend to have a high average rating. Moreover, the OLS specification is unable to rule out the effect of a movie's intrinsic quality on its box office revenues, which could be a confounding factor related to its release window. As such, the late effect of product reviews may be underestimated.

Hu et al. (2008) and Cui et al. (2012) examine the temporal change of the sales impact of product reviews on Amazon.com. Hu et al. (2008) split the data into early versus late periods and conducted a cross-tabulation comparison. How they split the PLC may be arbitrary. Also, the cross-tabulation approach may suffer from many uncontrolled factors. Cui et al. (2012) use review volume as the proxy of the PLC and observe an inverted U-shaped relationship between this variable and sales. However, review volume may not be an apt measure of PLC: a popular product may gather many reviews in a few days, while a niche product may only get a few comments during its entire life cycle.

Our study has several qualitative differences from the works mentioned above. First, we apply the fixed effects specification to the panel data to mitigate the concern that time-invariant movie features would bias the estimate. Second, we adopt a Heckman-style two-stage model (Heckman, 1979) to deal with the incidental selection issue so that our estimates would not suffer from the unbalanced panel due to the missing data points of movies with a shorter life cycle. Third, instead of splitting the sample by time or creating time dummies, we use a continuous variable – the number of days since release – to indicate the PLC. By examining the interaction between this measure and product review features, we investigate the continuous evolution of their impact on consumer demand over time.

Apart from these studies, the research by Gopinath et al. (2014) is perhaps the most similar to ours. In their paper, the authors have analyzed the impact of three types of online WOM–attribute-oriented, emotion-oriented, and recommendation-oriented WOM–on sales of cellular phones. The empirical results suggest that only the valence of recommendation-oriented WOM can positively affect sales, and such an effect increases over time. We extend this study of WOM on material goods (i.e., cellphones) by 1)

analyzing the economic role of online WOM across two PLCs (discussed below) and 2) comparing their impact on the consumer demand for experience goods (i.e., movies) produced in different cultures–Chinese and American movies, detailed in Sect. 2.2.

A novel perspective of our study is that we compare the role of online product reviews on another time dimension: when the movies are released (early) in theaters versus (late) online. This cross-stage comparison allows us to explore whether consumers use product reviews differently in the online market versus the local market and across platforms (i.e., cinema versus video streaming sites). Although Moon et al. (2010) also studied online movie reviews in two different markets (i.e., in the cinema vs. on Netflix), their analyses in the second stage do not address the impact of the movie reviews on consumer demand (probably due to the lack of such data). We fill this research gap in our study.

2.2 Foreign products and cultural gap

Another topic we cover is the differential role of reviews for products produced in a familiar culture versus foreign products. Specifically, in the context of the Chinese market, we compare the effect of reviews on consumer demand between domestically produced movies and American movies. To Chinese consumers, American movies are produced in a foreign culture, which is likely to have greater product uncertainty in contrast to domestically produced movies.

A cultural gap usually occurs when a disconnection is evident between the home and foreign markets (Hoskins & Mirus, 1998). Consumers buying products produced in other cultures may experience higher uncertainty (Safari et al., 2013). They need more information to appreciate a foreign product due to their lack of cultural background (Lee, 2006). For example, Gao et al. (2020) reveal that the informativeness of the movie title translation (which helps consumers understand the story) significantly improves a Hollywood movie's Chinese box office revenue.

Online product reviews should be another valuable information source to reduce such uncertainty stemming from cultural gaps. A few research works have compared consumers' review generation behaviors between cultures (e.g., Fang et al., 2013; Koh et al., 2010). Yet, to the best of our knowledge, little extant research analyzes how movie reviews affect consumers in selecting foreign products. The estimations in our study address this question.

2.3 Cross-platform release strategy

The last stream of research related to our study is the cross-platform release strategy, a popular topic in the marketing field. While Calzada and Valletti (2012) suggest the benefit of a shorter release gap, their models allow consumers to make repeat purchases; that is, they can watch movies both in the theater and online. Similarly, August et al. (2015) demonstrate that a shorter release gap can increase profits for high-quality movies. Luan and Sudhir (2022) identify a trade-off between the negative impact of a short gap and the dissipation of positive WOM as the gap increases. All of these studies mainly employ analytical modeling approaches.

Our empirical probe focuses on first-time buyers and accounts for the impact of online product reviews across the stages. We unveil the role of the release gap from the standpoint of consumers' utilization of product reviews. To some extent, the results complement the findings by Luan and Sudhir (2022): movies with good product reviews should shorten the release gap if they want to benefit from WOM continuously. By comparison, those not receiving positive reviews during the theatrical run should consider setting a longer gap between the two stages.

3 Hypotheses development

We develop the hypotheses for the temporal role of online product reviews in terms of three dimensions: their impact on consumer demand during the first (cinema release) stage, their impact on foreign cultural products in contrast to domestically produced products, and their effect on consumer demand during the late (online) stage.

3.1 The Role of product reviews within a product life cycle

Product life cycle (PLC) refers to the chronological stages of a product (i.e., introduction, growth, maturity, and decline) based on its sales and the movement of consumer groups (Levitt, 1965). The theory of diffusion of innovations (Rogers, 1962) provides a framework to explain further how consumer groups change during a product's life cycle. The term "innovation" in this theory can refer to any object perceived as new to consumers, including newly released products.

Throughout the diffusion process, five consumer groups sequentially adopt a product: innovators, early adopters, early majority, late majority, and laggards. The first two groups tend to be less risk-averse. Innovators are especially risk-tolerant and may adopt a new product even if there is insufficient information available about its performance. In addition, both innovators and early adopters have strong opinion leadership. They are inclined to give suggestions to influence future consumers. By contrast, the remaining three consumer groups (i.e., early and late majorities and laggards) are more conservative and skeptical during their purchase decision-making process. They seldom play the role of opinion leaders and tend to rely on early consumers' WOM. Although the consumer groups have no clear boundaries, they often appear chronologically, overlapping with a product's sales growth in its life cycle.

Because online reviews are often accumulated and recorded on websites, prospective consumers can readily find review information on a given website. In line with the characteristics of the first two consumer groups, we should observe that the sales impact of product reviews at an early period of product release is limited because these buyers are more risk-tolerant and do not necessarily need to rely on others' opinions. Note that although the level of product uncertainty at the early period of a PLC is very high (probably due to the lack of online WOM), innovators and early adopters may evaluate a product using other information,

e.g., existing product features. In our research setting, for example, early consumer groups may include fans of a movie star or lovers of a sci-fi series. Their deep interest in the product or its related characteristics may offset the uncertainty about the product or help them formulate a positive bias toward the product (Li & Hitt, 2008). As a result, they do not consult product reviews prior to making a purchase decision. By contrast, late consumer groups do not have such a strong tie to the product and tend to be more risk-averse, according to the theory of diffusion of innovation. They may not be persuaded to adopt a product unless it gathers enough consumer endorsements (i.e., positive WOM). As the PLC evolves, the later the consumer group, the more likely they are to refer to the existing buyers' opinions to make purchases. Hence, we should observe that the impact of product reviews on consumer demand increases throughout the PLC.

Our study aims to examine the sales impact of two review measures: review valence (i.e., the average rating) and review volume (i.e., the total number of reviews). Because most review platforms display and even highlight these two summarized statistics on the product page, consumers may use these two signals to decide whether to purchase the product. We formulate the hypotheses regarding these two dimensions of product reviews.

The average rating is a commonly used review feature in the literature, and several studies show evidence of it positively affecting sales (e.g., Chintagunta et al., 2010; Zhu & Zhang, 2010). From the consumers' perspective, such a statistic represents the overall valence of extant buyers' opinions of the product. A positive overall rating suggests that most existing buyers who leave a review have had an enjoyable product experience. Thus, it should encourage future buyers to adopt the product. Note that even if the volume of reviews is not large, a high valence may still indicate that the product is favored by a specific group of consumers, i.e., a niche product. For example, documentary movies are usually not a very popular genre (in contrast to, e.g., comedy, drama) and thus may not receive many reviews. However, a positively rated documentary movie can still be recognized as having good quality and thus attract moviegoers interested in this genre. Because late consumer groups are more reliant on the opinions of early buyers, we expect that the positive sales impact of review valence will increase over time. Formally:

H1a: During the cinema release stage, the impact of review valence on consumer demand increases over time.

Consumers also observe the total number of reviews on a typical product review website. Yet, such a statistic may not have a clear relationship with sales. Although review volume may indicate product awareness (Liu, 2006), a large number of reviewers criticizing the product may discourage future buyers from purchasing the product. Furthermore, when a product's sales growth reaches its peak and starts to decline, the cumulated review volume may be negatively tied to sales (Cui et al., 2012). Hence, we do not hypothesize a specific effect of review volume on consumer demand.

A higher review volume will become more meaningful if it is considered in relation to the valence of the opinions. In particular, a positive valence with a large number of reviews indicates that the product not only has good WOM but is also widely accepted by existing buyers. As such, in addition to the sales impact of review valence, the interaction between review valence and volume should have a positive effect on sales. Since consumer groups in late periods of a PLC tend to be more conservative and risk-averse, they may put greater weight on this signal to make purchases as well. This may be especially the case when consumers have a higher level of product uncertainty when evaluating foreign products, a point we will discuss in the ensuing section. In line with these arguments, we expect an increasingly positive effect of the interaction between review valence and volume on consumer demand over time.

H1b: During the cinema release stage, the impact of the interaction between review valence and review volume on consumer demand increases over time.

3.2 Domestically produced versus foreign cultural products

When conducting a pre-purchase evaluation of a foreign product, online product reviews may be a more crucial resource for potential consumers. A cultural gap–a disconnection between the home and foreign markets (Hoskins & Mirus, 1998) – may occur in this scenario. Buyers of products produced in other cultures may experience greater uncertainty (Safari et al., 2013) due to the lack of cultural background and knowledge to understand foreign products (Lee, 2006). Such uncertainty could have a notably negative impact on sales in Asian countries, where uncertainty avoidance is strong (Pratesi et al., 2021). In this situation, WOM from other buyers may help mitigate the concerns.

To a typical Chinese moviegoer, an American movie is produced in a distant culture. It is difficult for the Chinese audience to understand the language, the punchlines, the general setting of the story, and so on. According to Rogers (1962), innovators and early adopters usually have a higher social status and education. In our research setting, they are likely to be those familiar with American culture, including individuals with a college degree (English is a compulsory subject in Chinese universities) and those who have studied abroad in English-speaking countries, especially in the US. Similar to the early consumer groups of domestic movies, these individuals are less likely to be affected by product reviews since they have an understanding of (or even a preference for) foreign cultural products.

By contrast, those in the late periods will likely have insufficient knowledge to appreciate these products. Unlike domestically produced movies, early consumers' opinions about American movies may not be successfully accepted by late adopters if these reviews have a high valence but insufficient volume. Popular social topics in American culture, such as feminism (e.g., Down et al., 2020), race issues (e.g., Weaver, 2011), and democracy (e.g., Settich, 2013), which are often reflected in movies, are not typically part of Chinese cultural ideology. Even if the early consumer groups can understand these topics (since they tend to be familiar with the

culture), the messages in these movies are not entirely understandable to the late consumers. As a result, a high review valence is not persuasive enough.

Such a cultural gap is not a unique phenomenon in the Chinese market. In 2023, the hit Japanese animated movie Suzume (7.6/10 on IMDb) was released in the US. It was shortly off many theaters due to a moderate box office performance and limited WOM. In sharp contrast, the Hollywood production Super Mario Bros. Movie, which was released during the same period, has significantly outperformed Suzume, although it has a lower review valence (7.0/10 on IMDb). One of the reasons cited by the news articles is that American audiences can hardly understand the Japanese culture of self-sacrifice and conformity reflected in Suzume (Davis, 2023).

On the other hand, if a foreign movie obtains both high review valence and high review volume, it means that the majority of moviegoers widely accept the product. Such a movie may convey a core value that is readily acceptable to another culture. In our dataset, for example, the American animation Frozen has grown in considerable popularity in China. The power of true love between the two main characters and the importance of courage are not culture-specific and are thus appreciated by late Chinese consumer groups as well. This movie not only received a high average rating but also gathered a substantial review volume, which might be a major reason why it eventually grossed $48.2 million in China, beating all Chinese counterparts on the market. Taking all the above discussions into account, we posit that the interaction between review valence and volume will benefit foreign movies more in the long run.

H2: The impact of the interaction between review valence and volume on consumer demand for foreign movies is stronger than that of domestically produced movies, and such an effect increases over time.

3.3 The change in the role of product reviews across stages

After a movie's theatrical run, it is usually released on video streaming platforms. This unique feature of movies creates a challenge when comparing the role of online reviews across the two stages: whether we should treat the two stages as separate life cycles or a single one, and depending on the answer, different expectations of the role of product reviews in the second market (stage) will arise. We find arguments from both sides.

On the one hand, the earliest consumers during the online stage are not necessarily innovators or early adopters because the movie is no longer a brand-new product. As such, they may not have the characteristics of the early consumer groups during the cinema release stage. Considering that they prefer to wait to watch the movie online at a lower cost (a point we will discuss subsequently), these moviegoers may be more conservative than the late adopters in the theatrical run. In other words, although the product is released on a different platform in the second stage, its sales growth can be regarded as a continuous process from the previous stage, as it is exposed to the next consumer group. Second, these consumers may follow the movie's WOM during its theatrical run. By the time a movie is released online, it

should have garnered enough reviews to indicate the "true quality" of the product. Accordingly, the impact of online reviews on consumer demand might be even more exaggerated in the online stage.

On the other hand, compared with moviegoers in the cinema, online users' cost of watching an unfavorable movie is lower. They only need to make a monthly payment to a video streaming platform to get access to all movies on the site, and such fees are often lower than a single movie ticket. In our research setting, a typical monthly video streaming platform subscription costs users 15 Chinese yuan (CNY). By contrast, the average ticket price of a single movie being screened in the cinema is about 33 CNY, more than double the monthly payment.[3] Moreover, unlike the cinema viewing experience, online users can stop video streaming at any time if they do not enjoy the movie. With a lower consumption cost, consumers in the second stage may be more tolerant of movies with a lower review valence. Besides, consumers in the online stage make choices among a vast number of movies, in contrast to the limited number of movies screened in the cinema. Given these various factors, consumers' decisions may be less likely to be affected by reviews at this stage. Besides using reviews, they may adopt more heuristic rules (Payne et al., 1993), such as choosing movies based on their genre preference and relying on recommendations made by video streaming platforms. In line with these logical inferences, we expect a less significant effect of reviews on consumer demand.

Overall, the impact of product reviews on consumer demand after a movie's online release depends on the characteristics of consumers across stages – specifically, whether the later stage builds upon the earlier one or if they differ qualitatively. As such, we propose two alternative hypotheses.

H3a: The impact of online product reviews on consumer demand becomes stronger in the online stage if it is a continuous process of the cinema release stage.
H3b: The impact of online product reviews on consumer demand becomes less significant in the online stage if it is not a continuous process of the cinema release stage, i.e., if consumers adopt different product selection criteria.

4 Data and model

We conduct our study in the Chinese market. Since 2012, China has become the world's second-largest movie market in terms of box office revenues and is a key market for the Hollywood industry (BBC News, 2013), which offers an apt context to (1) study the evolution of consumer groups and (2) compare the role of reviews in affecting culturally different products. We gathered the product reviews from Douban Movie and the box office data from Maoyan.

[3] During our data collection, there was no option to purchase a single newly released movie on any streaming platform in China. Thus, we do not compare the price of watching a single movie in the theater with that of watching it online.

4.1 Movie review data

Douban Movie is a subdomain of Douban, a popular Chinese online social networking site where users read and share reviews of books, music, and movies (Davis, 2021). The iPhone App Store awarded Douban Movie the most influential movie community in China (Erdnase, 2013), implying its substantial impact on Chinese moviegoers' decision-making.

In 2014, we followed the cinema release schedule to collect product information on Douban for all movies released in China between January 18 and September 16. The data include a unique identifier of each movie, title, year of production, director(s), writer(s), cast, genre, country or region of production, language(s), release date, and movie length. We also took a snapshot of each movie's main page every 24 h to record the daily variation of product reviews, including the average rating, the total number of reviews, and the distribution of ratings. Panel A in Fig. 1 shows an example – a snapshot of *The Truth about Beauty* on April 5, 2014, translated into English.

On each product page, users can report that they have watched the movie by clicking the "watched" button. They can also add the movie to the list if they plan to watch it later. The website then updates the *total number of users who have watched the movie* and the *total number of users who have added the movie to the watchlist*. Panel B in Fig. 1 presents such information from the snapshot of the movie in Panel A. We use the first number to compute the daily user-reported viewership – a measure that proxies consumer demand.

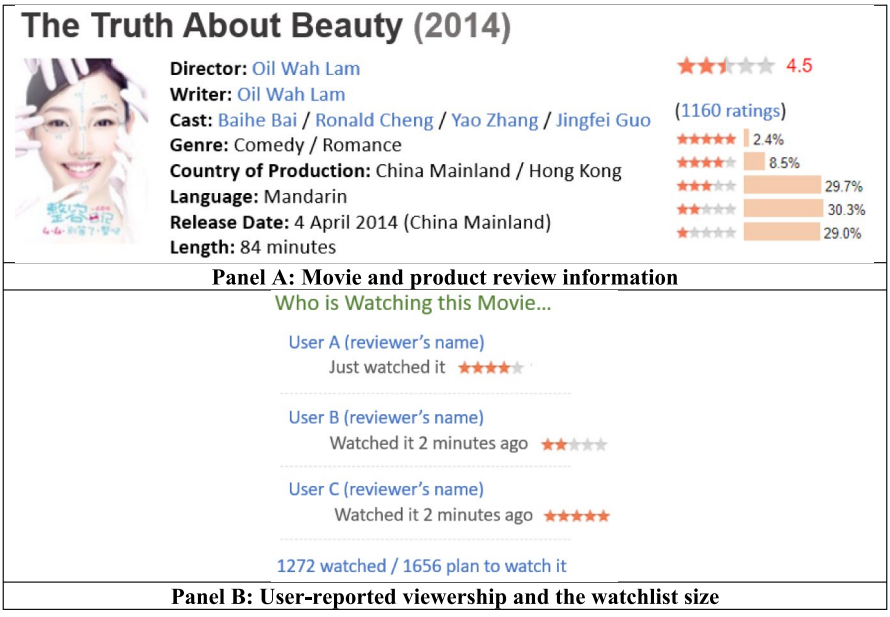


Fig. 1 Screenshots of a movie's webpage on Douban. Notes: (1) The average rating of 4.5 for the movie in **A** is on a scale of 10. (2) For privacy concerns, we do not reveal the names of the three individual reviewers displayed in **B**

4.2 Box office data and release information

For the cinema release stage analyses, we obtained the box office data to complement and validate the demand analyses using user-reported viewership. Specifically, we scraped the data from piaofang.maoyan.com, a public database owned by Maoyan that records the box office for movies released in China. This data set contains an identifier for each movie, daily box office revenues, average ticket price, and the number of screens. We matched the movies in this data set with those from the review data by checking the movie's title and release date.

To build the panel for the cinema release stage, we used each movie's release date as the beginning of its theatrical run and the last box office report date as the end of its release window. For online release, whenever a movie is released on popular Chinese video streaming platforms such as youku.com and iqiyi.com, Douban places the links to these sites on the movie's page. We set the earliest online release date for each movie as the start of its second stage. The original matched sample has 158 movies, including Chinese productions, American movies, and movies from other countries. To make the estimated results comparable across stages, we excluded movies not released online before the end of our data collection. This process led to a final sample of 47 domestically produced movies (i.e., the primary country of production is mainland China) and 17 American movies (i.e., the US is a major production country).

Figure 2 plots the consumer demand using an example movie, *The Breakup Guru*, released on June 27, 2014; its last box office report was on August 9, 2014. The y-axis denotes the demand (left scale: user-reported viewership; right scale: box office revenue), and the x-axis is the calendar date. The dashed line represents the box office revenue, which is very similar to the viewership denoted by the solid line, suggesting consistency between the two measures.

We also observe that the viewership dropped to a minimum level after the last box office report date (denoted by the dashed vertical line). This number rebounded eleven days later when the movie was released online on August 20, 2014 (denoted by the solid vertical line). The positive viewership between the two dates could be

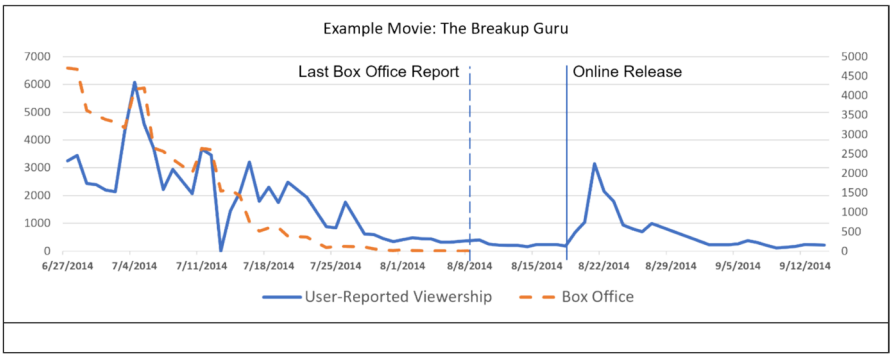


Fig. 2 An illustration of consumer demand across two release stages

the result of delayed self-reports by individual Douban users or a minimal number of users who live outside of mainland China but have access to the movie in their markets. In general, these patterns align with the movie's release schedule.

4.3 Main model

For movie i on day t, we model its consumer demand using the following equation.

$$\begin{aligned} log(Demand_{i,t}) = {} & \beta_1(RevValence_{i,t-1} - r) + \gamma_1(RevVolume_{i,t-1} - v) \\ & + \theta_1(RevValence_{i,t-1} - r)(RevVolume_{i,t-1} - v) + \lambda Age_{i,t} \\ & + \beta_2(Age_{i,t} - a)(RevValence_{i,t-1} - r) + \gamma_2(Age_{i,t} - a)(RevVolume_{i,t-1} - v) \\ & + \theta_2(Agei, t - a)(RevValence_{i,t-1} - r)(RevVolume_{i,t-1} - v) \\ & + \eta BaiduTrends_{i,t} + \varphi Weekend_{i,t} + \phi Holiday_{i,t} + \omega_i + \tau_t + \varepsilon_{i,t}, \end{aligned} \tag{1}$$

where *Demand* represents the dependent variable. We use user-reported viewership as the outcome in both the cinema release and online stages, operationalized as the number of Douban users who indicate they have watched the movie.

On the right-hand side of the equation, $RevValence - r$, $RevVolume - v$, and $(RevValence - r)\ (RevVolume - v)$ stand for the mean-centered review valence (i.e., the average score of all ratings), review volume (i.e., the total number of ratings), and their interaction, respectively. If we directly interact *RevValence* with *RevVolume*, the estimate β_1 will represent the partial effect of review valence on the viewership when there is no review (i.e., $RevVolume - 0$), and vice versa (Wooldridge, 2012), which is not practically meaningful. Thus, we mean-center the two variables in the interaction by abstracting each of them from its sample average, denoted by r and v.

To explore how product reviews' impact on consumer demand changes over time, we include the term *Age* – the number of days since release–and its interaction with *RevValence, RevVolume*, and their interaction. By setting the value of a in $(Age - a)$ as 1, 15, and 30, we can compare the partial effect of product reviews on the opening day, day 15, and day 30. We set the maximum value as 30 because this is the typical theatrical window in China (Toutiao, 2021).

In addition to the review measures, we include *Baidu Trends,* the search volume index on Baidu–the Chinese equivalent of Google–to indicate consumers' product interest due to other marketing activities, similar to Moe and Schweidel's (2012) usage of Google Trends. We also follow Duan et al. (2008) to control two indicators, *Weekend* and *Holiday*, because, logically, these factors should affect the outcome. During the cinema release stage, we account for an additional factor (not shown in Eq. 1): *Competing Movies*, i.e., the weighted number of movies being screened in the cinema that have the same genre. The weighted number is calculated based on the overlap of multiple genres between the focal movies and other movies on the market. Also, we use a dummy variable–*Released Online*–to indicate whether the movie has been released online. For the online stage estimations, we include the counterpart measure–*Being Screened*, which indicates that the movie is still being screened in the cinema.

We employ the movie-level fixed effects denoted by ω and calendar week dummies (i.e., which week the observation belongs to in 2014) to account for any time-invariant factors at the movie level (e.g., movie genre, production budget) and any specific events in a given week (e.g., any public events distracting consumers from watching the movie) that may have impacted the demand of a movie. We also include monthly dummies (i.e., January, February, etc.) to rule out the seasonal effect (e.g., the box office is usually higher in the summer). Finally, like Duan et al. (2008), we set a one-day lag between the product review measures and the dependent variable.

For the cinema release stage, we also regress the box office revenue on the same set of explanatory variables in Eq. 1 and two additional covariates – average ticket price (*Avg. Ticket*) and the number of screens (*Screens*) – to complement and validate the viewership estimations. Table 2 details the explanations and descriptive statistics for the variables in our estimations. Because we aim to compare the estimates between two types of cultural products (Chinese versus American movies) and across stages (theatrical run versus online release), we report the statistics based on this 2×2 dimension. We detail the samples for each analysis in the following section.

5 Main results

5.1 Cinema release stage

Based on the release date and the last box office report date, we obtained a sample of 47 Chinese movies with 1,417 observations and a sample of 17 American movies with 533 instances. These samples are unbalanced because the theatrical window differs across movies, depending on their box office performance. In addition, Maoyan does not report the daily box office for a movie if its revenue is below 1,000 CNY. As a result, we have smaller samples for the box office estimations: 1,028 observations for Chinese movies and 420 observations for American movies.

5.1.1 Selection model

The early finish of a movie's theatrical run (due to poor sales performance) and missing data points on low box office revenues lead to an incidental selection issue. If we do not account for this problem and directly use the truncated sample to run the consumer demand analyses, we may get biased estimates of the explanatory variables. To address this concern, we apply the two-stage estimation procedure proposed by Heckman (1979).

Specifically, in the first stage, we build a larger panel to run a selection model and analyze whether movie i has a box office record on day t. The State Administration of Press, Publication, Radio, Film, and Television in China usually sets a 30-day theatrical window for a newly released movie by encrypting the film through the

Table 2 Descriptive statistics of the key variables across samples

Variables	**Explanations (For movie *i* on day *t*)**											
Viewership	Number of users reporting that they have watched movie *i* on day *t*											
RevValence	Average score of all ratings for movie *i* on day *t* (on a scale of 10)											
RevVolume	Total number of ratings of movie *i* receives up until day *t*											
Age	Number of days since the release date (cinema release and online release) for movie *i*											
Baidu Trends	Baidu search index for movie *i* on day *t* (i.e., a proxy of the number of internet users searching for the movie title)											
Weekend	Whether day *t* is during the weekend (i.e., Friday, Saturday, or Sunday)											
Holiday	Whether day *t* is a public holiday in China											
Box Office Revenue	The box office revenue (in 1,000 CNY) of movie *i* on day *t*											
Avg. Ticket	The average ticket price (in CNY) of movie *i* on day *t*											
Competing Movies	Weighted number of movies being screened in the cinema that have the same genre on day *t*											
Screens	Total number of thousands of screens assigned by cinema lines to movie *i* on day *t*											
Released Online	Whether movie *i* has already been released online on day *t*											
Being Screened	Whether movie *i* is simultaneously being screened in the cinema on day *t*											
Descriptive Statistics	**Domestically Produced Chinese Movies (*n* = 47)**						**Imported American Movies (*n* = 17)**					
	Cinema Release Stage (*N* = 1,417)			**Online Stage (*N* = 2,973)**			**Cinema Release Stage (*N* = 533)**			**Online Stage (*N* = 905)**		
	Min	**Max**	**Mean**	**Min**	**Max**	**Mean**	**Min**	**Max**	**Mean**	**Min**	**Max**	**Mean**
Viewership	0	12,050	522.06	0	6,353	123.89	0	11,346	910.78	0	1913	113.92
RevValence	2.1	7.8	4.71	2.2	7.7	4.28	5.4	8.5	6.94	5.4	8.4	6.73
RevVolume	24	44,630	8,565.82	24	52,240	9,076.26	502	61,024	13,276.21	1,752	74,583	18,151.99
Age	1	60	18.49	1	90	39.95	1	54	17.97	1	90	40.86
Baidu Trends	0	506,400	14,407.04	0	132,193	5,825.84	66	187,431	24,729.52	0	140,994	7,353.93
Weekend	0	1	0.47	0	1	0.44	0	1	0.46	0	1	0.44
Holiday	0	1	0.07	0	1	0.03	0	1	0.05	0	1	0.03
Box Office	2	98,141	4,769.14	–	–	–	5	93,512	7,057.01	–	–	–

Table 2 (continued)

Avg. Ticket	17.2	207	32.62	–	–	–	22	65.4	35.17	–	–	–
Competing Movies	0	20	6.62	–	–	–	0	20	8.44	–	–	–
Screens	0.002	43	6.03	–	–	–	0.032	37	7.46	–	–	–
Released Online	0	1	0.12	–	–	–	0	1	0.06	–	–	–
Being Screened	–	–	–	0	1	0.06	–	–	–	0	1	0.04

(1) Following the literature (e.g., Chintagunta et al., 2010), we take the natural log of *Box Office*, *Viewership*, *RevValence*, *RevVolume*, and *Baidu Trends*. (2) The samples for the box office analyses (N = 1,028 for Chinese movies; N = 420 for American movies) are smaller than those for the viewership estimations because Maoyan does not report box office revenues below 1,000 CNY. The statistics of the box office revenue, average ticket price, and screens are computed using these smaller samples

Table 3 The probit estimation of the box office record

Dependent Variable	*Box Office Record*
RevValence	1.0477 (1.7937)
RevVolume	−1.8759* (1.0028)
RevValence × *RevVolume*	–1.1178 (1.3483)
Age	–0.1833*** (0.0478)
Baidu Trends	0.8254** (0.3706)
Weekend	–0.0362 (0.1603)
Holiday	0.3952 (0.3341)
Released Online	–0.2429 (0.7553)
Box Office Report Rate by the Distributors	1.5135** (0.7501)
Movie Fixed Effects	Yes
N	2,289

Clustered standard errors at the movie level are shown in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

key delivery message (KDM)[4] (Toutiao, 2021). A film can extend its release time for another 15 or 30 days, depending on its box office performance. Alternatively, a cinema can end a movie's theatrical run before the expiration date of the KDM if it does not generate enough sales. Using this rule, we set a panel that uses 30-day, 45-day, and 60-day observation periods for each movie based on its actual box office records. If a movie's last box office date is within 30 days after its release, we set the observation period as 30 days; if its box office report period is between 30 and 45 days, we set the observation period as 45 days; for the remaining movies in our panel, we set 60 days as their theatrical window.

We then regress the binary outcome – whether movie i has a box office record on day t – on the following variables in Eq. 1: *RevValence* − *r*, *RevVolume* − *v*, *(RevValence* − *r) (RevVolume* − *v)*, *Age*, *Baidu Trends*, *Weekend*, *Holiday*, and *Released Online*. We adopted a movie-level fixed effects specification in a probit framework following Heckman (1979). To meet the restriction exclusion in the selection model, we included a variable exogenous to the consumer demand estimations. In particular, we calculated the *Box Office Report Rate by the Distributors*, which is the fraction of all movies by the same distributors outside the release window of the focal

[4] For further information about the KDM message, refer to https://cinepedia.com/security/key-delivery-message/

movie with a box office record on day *t*. Note that the movies we used in computing this variable exclude the focal movie. They are the other movies distributed by the same distributor. Also, we only consider movies being screened outside of the theatrical run of the focal movie to strengthen the exclusion restrictions. The logic is that a distributor's negotiation power can affect a movie's cinema screen allocation and, thus, the box office report. Whether a movie has a box office record on day *t* should be positively associated with the same outcome of other movies (on day *t*) by the same distributor. However, this factor should not affect the focal movie's exact box office revenue – the outcome at the second stage.

Table 3 reports the results of the selection model using all 64 movies during the cinema release stage. The estimates indicate that *Age* and *RevVolume* can negatively affect the outcome. Such findings are reasonable, given that a movie is more likely to end its life cycle in a later period or when it receives more cumulative product reviews. On the other hand, higher *Baidu Trends* imply a higher level of popularity for the movie, thus helping extend its theatrical window, i.e., with a greater chance of getting a box office record on day *t*. Most importantly, the variable *Box Office Report Rate by the Distributors* has a positive and significant estimate, suggesting the impact of the distributors' negotiation power on the box office record of the focal movie on a specific day. We have also run a consumer demand model that includes the *Box Office Report Rate by the Distributors*. The results reveal no significant impact of this variable on the box office revenue or the viewership, further supporting the exclusion restriction assumption.

Next, we take the inverse Mills ratio (*IMR*) from the selection model and include it in the second stage, i.e., the demand estimations. Including the *IMR* of the first-stage residuals in the models at the second stage helps mitigate the concern that the incidental selection issue would cause biased estimates (Bushway et al., 2007; Heckman, 1979).

5.1.2 Domestically produced Chinese movies

Table 4 reports the consumer demand estimations of Chinese movies. The outcomes are user-reported viewership in columns 1–4 and box office revenues in columns 5–8. The first model in each set of estimations is the baseline model, in which we do not include any interactions.

Across all models, we observe a positive and significant coefficient of *RevValence*, consistent with previous research's finding that the valence of reviews affects box office revenues (e.g., Chintagunta et al., 2010). Moreover, the interaction between *RevValence* and *RevVolume* can positively affect consumer demand in most models. This result is in line with our expectations–movies favored by a substantial number of existing buyers tend to have better sales performance. The negative coefficient of *RevVolume* in the viewership models is also reasonable. Because we mean-centered *RevValence* and *RevVolume* using the average value of the entire cinema release sample ($r = 5.3$, $v = 9{,}853.33$), an increase in the number of reviews when the average rating is only 5.3 out of 10 may only decrease the viewership.

Our focus is on the interaction between *Age* and review measures. First, we obtain a positive and significant coefficient of *Age* × *RevValence* in the viewership

Table 4 The effect of product reviews on consumer demand for Chinese movies at the cinema release stage

	User-reported viewership				*Box office revenue*			
	Baseline	Age = 1	Age = 15	Age = 30	Baseline	Age = 1	Age = 15	Age = 30
RevValence	3.8462***	2.7613**	3.7939***	4.9003***	1.7451***	1.8633***	2.3242***	2.8181***
	(1.0547)	(1.1082)	(0.9913)	(0.9543)	(0.4805)	(0.5738)	(0.4264)	(0.6020)
RevVolume	–0.6058***	–0.5363***	–0.5615***	–0.5886***	0.1272	0.2294**	0.1338	0.0313
	(0.1261)	(0.1521)	(0.1499)	(0.1604)	(0.1125)	(0.1026)	(0.1021)	(0.1363)
RevValence × RevVolume		0.2587	0.5551**	0.8726***		0.5772***	0.7628***	0.9616***
		(0.3035)	(0.2709)	(0.2835)		(0.2020)	(0.2247)	(0.3417)
Age	–0.1008***	–0.1073***	–0.1073***	–0.1073***	–0.2238***	–0.2320***	–0.2320***	–0.2320***
	(0.0233)	(0.0223)	(0.0223)	(0.0223)	(0.0183)	(0.0169)	(0.0169)	(0.0169)
Age × RevValence		0.0738***	0.0738***	0.0738***		0.0329	0.0329	0.0329
		(0.0210)	(0.0210)	(0.0210)		(0.0279)	(0.0279)	(0.0279)
Age × RevVolume		–0.0018	–0.0018	–0.0018		–0.0068	–0.0068	–0.0068
		(0.0030)	(0.0030)	(0.0030)		(0.0044)	(0.0044)	(0.0044)
Age × RevValence × RevVolume		0.0212***	0.0212***	0.0212***		0.0133	0.0133	0.0133
		(0.0079)	(0.0079)	(0.0079)		(0.0113)	(0.0113)	(0.0113)
Baidu Trends	0.3488**	0.3600***	0.3600***	0.3600***	0.1090	0.1665*	0.1665*	0.1665*
	(0.1396)	(0.1348)	(0.1348)	(0.1348)	(0.0822)	(0.0945)	(0.0945)	(0.0945)
Weekend	0.2466***	0.2640***	0.2640***	0.2640***	0.4822***	0.4928***	0.4928***	0.4928***
	(0.0827)	(0.0826)	(0.0826)	(0.0826)	(0.0900)	(0.0870)	(0.0870)	(0.0870)
Holiday	0.1488	0.1832	0.1832	0.1832	–0.1965	–0.1420	–0.1420	–0.1420
	(0.1132)	(0.1239)	(0.1239)	(0.1239)	(0.1293)	(0.1198)	(0.1198)	(0.1198)
Released Online	1.6976***	1.2976***	1.2976***	1.2976***	–0.2124	–0.3560	–0.3560	–0.3560
	(0.4065)	(0.3839)	(0.3839)	(0.3839)	(0.2612)	(0.2813)	(0.2813)	(0.2813)
Screens					0.0537***	0.0483***	0.0483***	0.0483***
					(0.0118)	(0.0130)	(0.0130)	(0.0130)
Average Ticket					0.0041	0.0066*	0.0066*	0.0066*
					(0.0035)	(0.0034)	(0.0034)	(0.0034)

Table 4 (continued)

	User-reported viewership				*Box office revenue*			
	Baseline	Age = 1	Age = 15	Age = 30	Baseline	Age = 1	Age = 15	Age = 30
Competing Movies	–0.0162 (0.2319)	–0.1584 (0.1935)	–0.1584 (0.1935)	–0.1584 (0.1935)	–0.2346 (0.2466)	–0.3010 (0.2276)	–0.3010 (0.2276)	–0.3010 (0.2276)
IMR	–0.1702 (0.1232)	–0.0289 (0.0850)	–0.0289 (0.0850)	–0.0289 (0.0850)	–0.4098 (0.2517)	–0.3353 (0.2119)	–0.3353 (0.2119)	–0.3353 (0.2119)
Movie-Level FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Week Dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month Dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	1,417	1,417	1,417	1,417	1,028	1,028	1,028	1,028
Adjusted R^2	0.8682	0.8789	0.8789	0.8789	0.9231	0.9289	0.9289	0.9289

Clustered standard errors at the movie level are shown in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

estimations, which indicates that the impact of review valence on consumer demand increases over time. Comparing the estimate of *RevValence* across the three models using $a = 1$, 15, and 30, respectively, we find that this number increases dramatically from 2.7613 on Day 1 to 4.9003 on Day 30. Second, we observe a significant and positive estimate of *Age* × *RevValence* × *RevVolume*. This finding suggests that movies with a higher review valence *and* volume will benefit more from product reviews over time. We find further evidence by comparing the coefficient of *RevValence* × *RevVolume* across columns 2 to 4. It is insignificant on the opening day but becomes significant in the latter two periods. In summary, these findings support H1a and H1b.

A potential concern of the estimations in the first four columns is that users who report their watching behavior on Douban may not represent the overall population of moviegoers in China. To mitigate such an issue, we run the models using the box office revenue as the dependent variable. The results in columns 5–8 are generally consistent with those in the viewership analyses. Although the coefficient of *Age* × *RevValence* is not significant, we observe a substantial increase in the estimate of *RevValence* from 1.8633 on Day 1 to 2.8181 on Day 30. Similarly, we find that the coefficient of *RevValence* × *RevVolume* has increased dramatically from 0.5772 on Day 1 to 0.9616 on Day 30. More importantly, we test the joint significance of the two interactions – *Age* × *RevValence* and *Age* × *RevValence* × *RevVolume* – and obtain an F statistic of 8.6994 (p-value < 0.001), which indicates that the two interactions matter. Although they do not have a direct impact on the box office revenues, they can jointly affect the outcome. Overall, the findings between the two sets of estimations are consistent, i.e., both of them reveal that review valence has an increasingly stronger impact on consumer demand across time, supporting H1a, and the interaction effect between review valence and volume on consumer demand increases over time, which supports H1b.

5.1.3 Imported American movies

Table 5 presents the estimation results using the sample of American movies. First, *RevValence* does not obtain a significant estimate across all models, including the baselines. As we discussed in Sect. 3.2, review valence itself can hardly persuade consumers to adopt a foreign product in two aspects. Early consumer groups do not necessarily need to rely on others' opinions since they have enough knowledge about the cultural background of the product and may have a preference for certain product characteristics (Li & Hitt, 2008). To the late consumer groups, it is hard for them to understand the ideology in American movies, even if the opinions of early adopters are positive.

By comparison, we obtain a significant and positive estimate of *Age* × *RevValence* × *RevVolume* across all models, indicating an increasingly positive impact of review valence *and* volume on consumer demand. This result is further supported by the increase in both the significance level and the magnitude of the estimate of *RevValence* × *RevVolume* from Day 1 to Day 30 in the two sets of demand estimations. Such a change suggests that only American movies receiving a substantial number of positive reviews may become successful in the Chinese market. Moreover, this

Table 5 The effect of product reviews on consumer demand for American movies at the cinema release stage

	User-reported viewership				Box office revenue			
	Baseline	Age = 1	Age = 15	Age = 30	Baseline	Age = 1	Age = 15	Age = 30
RevValence	2.9763 (5.3654)	6.2703 (4.4172)	6.1628 (3.8715)	6.0476 (3.8870)	–0.0893 (5.5359)	9.7031 (6.7207)	6.2556 (5.9343)	2.5618 (6.5477)
RevVolume	–0.1525 (0.1309)	–0.3742 (0.4905)	–1.5175*** (0.4485)	–2.7425*** (0.4564)	0.9396*** (0.1474)	1.2167* (0.6551)	0.1082 (0.6471)	–1.0795 (0.9244)
RevValence × RevVolume		0.8621 (1.2143)	3.0924** (1.3326)	5.4820*** (1.6889)		–0.0869 (1.5176)	3.3639* (1.9253)	7.0612** (3.1157)
Age	–0.0616*** (0.0218)	–0.0343 (0.0252)	–0.0343 (0.0252)	–0.0343 (0.0252)	–0.2422*** (0.0412)	–0.2166*** (0.0705)	–0.2166*** (0.0705)	–0.2166*** (0.0705)
Age × RevValence		–0.0077 (0.1069)	–0.0077 (0.1069)	–0.0077 (0.1069)		–0.2463 (0.2052)	–0.2463 (0.2052)	–0.2463 (0.2052)
Age × RevVolume		–0.0817*** (0.0107)	–0.0817*** (0.0107)	–0.0817*** (0.0107)		–0.0792** (0.0320)	–0.0792** (0.0320)	–0.0792** (0.0320)
Age × RevValence × RevVolume		0.1593*** (0.0416)	0.1593*** (0.0416)	0.1593*** (0.0416)		0.2465** (0.1017)	0.2465** (0.1017)	0.2465** (0.1017)
Baidu Trends	0.4784*** (0.1509)	0.4835*** (0.1651)	0.4835*** (0.1651)	0.4835*** (0.1651)	0.7471*** (0.2123)	0.6877** (0.2927)	0.6877** (0.2927)	0.6877** (0.2927)
Weekend	0.1862** (0.0891)	0.1536** (0.0648)	0.1536** (0.0648)	0.1536** (0.0648)	0.6231*** (0.1430)	0.5734*** (0.1483)	0.5734*** (0.1483)	0.5734*** (0.1483)
Holiday	–0.0740 (0.1206)	–0.1010 (0.1501)	–0.1010 (0.1501)	–0.1010 (0.1501)	0.3136 (0.3220)	0.2914 (0.3404)	0.2914 (0.3404)	0.2914 (0.3404)
Released Online	0.0037 (0.2850)	0.3072 (0.1957)	0.3072 (0.1957)	0.3072 (0.1957)	–0.5004** (0.2052)	–0.0131 (0.3666)	–0.0131 (0.3666)	–0.0131 (0.3666)
Screens					0.0520** (0.0226)	0.0364 (0.0236)	0.0364 (0.0236)	0.0364 (0.0236)
Average Ticket					–0.0212 (0.0179)	–0.0239 (0.0193)	–0.0239 (0.0193)	–0.0239 (0.0193)

Table 5 (continued)

	User-reported viewership				Box office revenue			
	Baseline	Age = 1	Age = 15	Age = 30	Baseline	Age = 1	Age = 15	Age = 30
Competing Movies	–0.1352 (0.2220)	0.0617 (0.1581)	0.0617 (0.1581)	0.0617 (0.1581)	–0.6314** (0.3178)	–0.1088 (0.4289)	–0.1088 (0.4289)	–0.1088 (0.4289)
IMR	–0.1102 (0.0921)	–0.1094 (0.1134)	–0.1094 (0.1134)	–0.1094 (0.1134)	0.0137 (0.2332)	0.0060 (0.2698)	0.0060 (0.2698)	0.0060 (0.2698)
Movie-Level FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Week Dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month Dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	533	533	533	533	420	420	420	420
Adjusted R^2	0.8528	0.8619	0.8619	0.8619	0.9070	0.9104	0.9104	0.9104

Clustered standard errors at the movie level are shown in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

interaction effect becomes increasingly important over time. Also, the coefficients of this interaction term have a much higher magnitude than the corresponding estimates in the models of Chinese movies, which is what we expected based on the discussions in Sect. 3.2. Overall, the results in these estimations support H2.

5.2 Online release stage

Finally, we explore the role of product reviews during the online release stage. In this stage, we use the user-reported viewership as the dependent variable. Because a movie is unlikely to be removed from a video streaming site, the end of the online life cycle has no clear definition. We find that the viewership of all movies drops to a minimum number with limited variations by Day 90. As such, we set the maximum number of days since the online release as 90 so that the window roughly covers the entire life cycle of the online stage. This criterion Helps extract 2,973 observations for the Chinese movies and 905 instances for the American movies. Table 6 reports the results at the online stage. Columns 1–4 show the estimates using the sample of Chinese movies, and columns 5–8 present the results of American movies.

Surprisingly, we obtain very different estimates for the two samples. For Chinese movies, the effect of *RevValence* generally becomes even stronger in the online stage. Its coefficient is 4.6734 on Day 1 and 4.7099 on Day 30. In addition, this effect increases over time, as indicated by the estimate of *Age*×*RevValence* being positive, albeit not significant. We observe a similar pattern regarding the interaction between *RevValence* and *RevVolume*; its impact continuously increases from the cinema release stage, and the magnitude of the coefficient is much stronger than that in the early stage. These findings suggest a continuous PLC, which seems to support H3a.

The estimations for American movies reveal a distinct pattern. First, the interaction between *RevValence* and *RevVolume* no longer affects the viewership. Second, we obtain a positive and significant estimate of *RevVolume*: Consumers seem to be more interested in movies with higher product awareness (as indicated by the review volume), regardless of the valence of early buyers' opinions. These results demonstrate a qualitatively different role of product reviews in the online stage, which seems to support H3b.

6 Mechanism exploration

6.1 Low product differentiation among American movies

What is the reason for the qualitatively different findings in the online stage between Chinese and American movies? We propose two potential explanations based on the data exploration of the two samples. First, we find that the average rating of American movies tends to be higher and has a lower variation than that of Chinese movies. The lowest rating for American movies at the online stage is 5.4; its median and average are 6.5 and 6.7, respectively, and the highest value is 8.4. The

Table 6 The effect of product reviews on consumer demand at the online stage

	Chinese movies				*American movies*			
	Baseline	Age = 1	Age = 15	Age = 30	Baseline	Age = 1	Age = 15	Age = 30
RevValence	0.9480 (0.8749)	4.6734*** (1.3646)	4.6910*** (1.3870)	4.7099*** (1.4135)	0.7717 (6.1282)	3.1054 (5.2815)	2.6891 (5.2774)	2.2728 (5.3741)
RevVolume	0.1704 (0.3869)	0.1683 (0.3950)	0.1604 (0.3960)	0.1519 (0.3972)	1.6441*** (0.5719)	5.6955** (2.7068)	5.7614** (2.7200)	5.8274* (2.7342)
RevValence × RevVolume		1.5202*** (0.5059)	1.6597*** (0.5178)	1.8092*** (0.5319)		–6.7146 (5.4492)	–6.6062 (5.4796)	–6.4978 (5.5178)
Age	–0.0057 (0.0113)	–0.0076 (0.0111)	–0.0076 (0.0111)	–0.0076 (0.0111)	–0.0302* (0.0173)	–0.0355*** (0.0131)	–0.0355*** (0.0131)	–0.0355*** (0.0131)
Age × RevValence		0.0013 (0.0044)	0.0013 (0.0044)	0.0013 (0.0044)		–0.0278 (0.0488)	–0.0278 (0.0488)	–0.0278 (0.0488)
Age × RevVolume		–0.0006 (0.0019)	–0.0006 (0.0019)	–0.0006 (0.0019)		0.0044 (0.0037)	0.0044 (0.0037)	0.0044 (0.0037)
Age × RevValence × RevVolume		0.0100*** (0.0020)	0.0100*** (0.0020)	0.0100*** (0.0020)		0.0072 (0.0139)	0.0072 (0.0139)	0.0072 (0.0139)
Baidu Trends	0.2459*** (0.0844)	0.2304*** (0.0787)	0.2304*** (0.0787)	0.2304*** (0.0787)	0.1545* (0.0904)	0.1608* (0.0868)	0.1608* (0.0868)	0.1608* (0.0868)
Weekend	0.0867* (0.0469)	0.0912** (0.0456)	0.0912** (0.0456)	0.0912** (0.0456)	0.2192*** (0.0548)	0.2117*** (0.0504)	0.2117*** (0.0504)	0.2117*** (0.0504)
Holiday	0.1993** (0.0973)	0.1935** (0.0945)	0.1935** (0.0945)	0.1935** (0.0945)	–0.1151 (0.1102)	–0.1176 (0.1128)	–0.1176 (0.1128)	–0.1176 (0.1128)
Being Screened	0.4401** (0.1725)	0.5657*** (0.1522)	0.5657*** (0.1522)	0.5657*** (0.1522)	–0.1165 (0.0983)	–0.1899* (0.1083)	–0.1899* (0.1083)	–0.1899* (0.1083)
Movie-Level FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Week Dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month Dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	2,973	2,973	2,973	2,973	905	905	905	905

Table 6 (continued)

	Chinese movies				*American movies*			
	Baseline	Age = 1	Age = 15	Age = 30	Baseline	Age = 1	Age = 15	Age = 30
Adjusted R^2	0.8755	0.8790	0.8790	0.8790	0.8471	0.8507	0.8507	0.8507

Clustered standard errors at the movie level are shown in parentheses.

*$p < 0.1$, **$p < 0.05$, ***$p < 0.01$

corresponding values for the Chinese movies are 2.2, 3.9, 4.3, and 7.7. Furthermore, the variation of review valence for Chinese movies (s.d. = 1.44) is much larger than that of American movies (s.d. = 0.87). Assuming consumers at the online stage use less restrictive product selection criteria (per discussions in Sect. 3.3), they may not use review valence to differentiate among American movies since their WOM is less differentiable.

If this is the case, we should obtain similar results in columns 6–8 of Table 6 using a sample of Chinese movies similar to the American products. We have to admit that there are other differences between the two sets of movies as well. To build a sample of Chinese movies that resemble their American counterparts, we apply the propensity score matching method. Notably, we match observations in the sample of Chinese movies with those of American movies using *RevValence*, *RevVolume*, *Baidu Trends*, *Weekend*, and *Holiday*. During the process, we use the nearest neighbor algorithm with a ratio = 3 (i.e., three most similar neighbors) and allow a Chinese movie to be matched with multiple American movies. The second choice facilitates more accurate matching, particularly when the control group has fewer observations that resemble the treatment units (Ho et al., 2007). The matching produces 266 observations of Chinese movies that are most similar to the 905 instances in the sample of American movies. T-tests show that, after matching, the average values of all covariates have no significant difference between the two groups.

The first three columns in Table 7 report the online stage estimation results using the matched sample. In general, we find patterns similar to those in columns 2 to 4 in Table 6. That is, the effects of *RevValence* and *RevValence*×*RevVolume* on self-reported viewership increase over time. In addition, the estimate of *Age*×*RevValence*×*RevVolume* is positive and significant, suggesting an increasing effect of the interaction between review valence and review volume on consumer demand. In summary, these findings do not support the first explanation.

6.2 Change of consumer groups due to longer release gaps

Our second exploration is based on the gap between the two release stages. We use the difference between the last box office report date and the online release date to indicate the cross-stage release gap. The descriptive statistics reveal that American movies have a much longer release gap (mean = 38.88 days) than their Chinese counterparts (mean = 8.68 days). More than 84.78% of Chinese movies in our data (i.e., 39 out of 47) were released within 30 days after the movie was out of the theater, whereas this number for American movies is only 41.18%.

We conjecture that the release gap is the potential driver for the different findings in Sect. 5.2. When a movie is released shortly after (or even before the end of) its theatrical run, the product remains available across stages. In this case, those planning to watch the movie online during the first stage will probably be the next consumer group. Because the movie was barely out of the theater, its online buzz is still evolving, and consumers' impressions of the products are affected by the continuous WOM (Luan & Sudhir, 2022). As such, the two stages may not be qualitatively

Table 7 Mechanism explorations using the sample of Chinese movies at the online stage

	Propensity Score Matched Sample			Release Gap × Online Reviews					
	Age = 1	Age = 15	Age = 30	OLS	No Gap	Gap = 15	Gap = 30	Gap = 45	Gap = 60
RevValence	12.9195	13.1309*	13.3574*	1.8039***	5.8351***	3.7522***	1.6693	−0.4137	−2.4966
	(8.2006)	(7.6782)	(7.2707)	(0.5584)	(1.1579)	(1.2294)	(1.7203)	(2.3843)	(3.1125)
RevVolume	−0.0487	−0.1774	−0.3153	0.7418***	−0.3677	0.2036**	0.7749***	1.3461***	1.9173***
	(0.6847)	(0.6427)	(0.6039)	(0.0887)	(0.2436)	(0.2437)	(0.2649)	(0.3030)	(0.3524)
RevValence × RevVolume	4.8559**	5.6548**	6.5108***	0.4920***	1.5626**	1.2195***	0.8764	0.5333	0.1902
	(2.3475)	(2.3293)	(2.3231)	(0.1766)	(0.4941)	(0.4525)	(0.7051)	(1.0589)	(1.4413)
Age	−0.0814*	−0.0814*	−0.0814*	−0.0116***	−0.0063	−0.0063	−0.0063	−0.0063	−0.0063
	(0.0434)	(0.0434)	(0.0434)	(0.0031)	(0.0109)	(0.0109)	(0.0109)	(0.0109)	(0.0109)
Age × RevValence	0.0151	0.0151	0.0151	0.0041	0.0001	0.0001	0.0001	0.0001	0.0001
	(0.0802)	(0.0802)	(0.0802)	(0.0040)	(0.0043)	(0.0043)	(0.0043)	(0.0043)	(0.0043)
Age × RevVolume	−0.0092*	−0.0092*	−0.0092*	0.0002	−0.0010*	−0.0010*	−0.0010*	−0.0010*	−0.0010*
	(0.0052)	(0.0052)	(0.0052)	(0.0009)	(0.0005)	(0.0005)	(0.0005)	(0.0005)	(0.0005)
Age × RevValence × RevVolume	0.0571***	0.0571***	0.0571***	0.0068***	0.0105***	0.0105***	0.0105***	0.0105***	0.0105***
	(0.0120)	(0.0120)	(0.0120)	(0.0019)	(0.0020)	(0.0020)	(0.0020)	(0.0020)	(0.0020)
Release Gap				0.0131**					
				(0.0061)					
Release Gap × RevValence				−0.0115	−0.1389***	−0.1389***	−0.1389***	−0.1389***	−0.1389***
				(0.0189)	(0.0533)	(0.0533)	(0.0533)	(0.0533)	(0.0533)
Release Gap × RevVolume				0.0014	0.0381***	0.0381***	0.0381***	0.0381***	0.0381***
				(0.0021)	(0.0049)	(0.0049)	(0.0049)	(0.0049)	(0.0049)
Release Gap × RevValence × RevVolume				−0.0081	−0.0229	−0.0229	−0.0229	−0.0229	−0.0229
				(0.0062)	(0.0272)	(0.0272)	(0.0272)	(0.0272)	(0.0272)
Baidu Trends	0.0247	0.0247	0.0247	0.1421**	0.2325***	0.2325***	0.2325***	0.2325***	0.2325***
	(0.0307)	(0.0307)	(0.0307)	(0.0582)	(0.0724)	(0.0724)	(0.0724)	(0.0724)	(0.0724)
Weekend	0.2264**	0.2264**	0.2264**	0.1110***	0.0805*	0.0805*	0.0805*	0.0805*	0.0805*
	(0.0940)	(0.0940)	(0.0940)	(0.0324)	(0.0464)	(0.0464)	(0.0464)	(0.0464)	(0.0464)

Table 7 (continued)

	Propensity Score Matched Sample			Release Gap × Online Reviews					
	Age = 1	Age = 15	Age = 30	OLS	No Gap	Gap = 15	Gap = 30	Gap = 45	Gap = 60
Holiday	0.1707 (0.3106)	0.1707 (0.3106)	0.1707 (0.3106)	0.1874* (0.1029)	0.1948** (0.0883)	0.1948** (0.0883)	0.1948** (0.0883)	0.1948** (0.0883)	0.1948** (0.0883)
Being Screened	0.6457** (0.2654)	0.6457** (0.2654)	0.6457** (0.2654)	0.6901** (0.2822)	0.2968** (0.1417)	0.2968** (0.1417)	0.2968** (0.1417)	0.2968** (0.1417)	0.2968** (0.1417)
Movie-Level FE	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes
Week Dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month Dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	266	266	266	2,973	2,973	2,973	2,973	2,973	2,973
Adjusted R^2	0.9188	0.9188	0.9188	0.7752	0.8822	0.8822	0.8822	0.8822	0.8822

Clustered standard errors at the movie level are shown in parentheses.

*$p < 0.1$, **$p < 0.05$, ***$p < 0.01$.

different, and the early adopters during the online stage can be regarded as the "later" adopters in the former stage.

By contrast, when the release gap is substantially large, the impact of online product reviews dissipates. Those planning to watch the movie during the cinema release stage may gradually lose interest due to the unavailability of the product. When the movie is released online, it might attract a new consumer group that differs from those in the early stage. On streaming platforms, consumers encounter a vast selection of movies (e.g., hundreds of movies), unlike the limited choices available in theaters. The reference for comparison differs significantly. With such an abundance of options, they may be less likely to rely on review scores when making decisions. Besides using reviews, they may use heuristic-based selection strategies (Payne et al., 1993). For example, they may choose movies with a higher review volume since it signals that many consumers discuss them frequently. Watching these movies can help them actively engage in conversations with friends or other online users (Godes & Mayzlin, 2004). Second, video streaming sites often recommend movies that online users frequently discuss. Consumers may be more likely to watch these movies based on the recommendations. As a result, we observe a significant effect of review volume on the viewership.

To test the above conjectures, we examine the interaction between the cross-stage release gap and product review measures, and include them in the models for Chinese movies in the online stage. Columns 4–9 in Table 7 report the results. First, since the release gap is time-invariant (which is ruled out by the movie-level fixed effects), we run an OLS model to obtain the direct impact of this variable on the outcome. As the estimate in column 4 suggests, the release gap has a positive effect on the viewership. Yet, such an impact is negligible (i.e., 0.0131) compared with the estimates of product review measures (e.g., 1.8039 for *RevValence* and 0.7418 for *RevVolume*). The following models include the movie-level fixed effects. The results produce a significant and negative coefficient of *Release Gap*×*RevValence*, meaning that the impact of review valence on consumer demand decreases as the gap between the two stages increases. When comparing the estimate of *RevValence* across the last five columns in Table 7, we find that it decreases and becomes insignificant in the last three models, i.e., when the release gap becomes longer than 30 days.

Second, although we do not obtain a significant coefficient of *Release Gap*×*RevValence*×*RevVolume*, we find that the effect of *RevValence*×*RevVolume* on viewership drops from column 5 to column 9 in Table 7. It becomes insignificant when the release gap reaches 30 days. On the opposite side, the estimate of *RevVolume* increases as the gap increases. In the last three columns in Table 7, the findings of online reviews are qualitatively the same as those using the sample of American movies at the online stage (i.e., columns 6 to 8 in Table 6). These results support the conjecture of the qualitative change in consumer behavior due to a more prolonged release gap.

We find further evidence for the above arguments using the second statistic illustrated in Fig. 1B. In particular, we derive the watchlist size–the total number of users who have added the movie to the list–for each movie. This number should increase if more users indicate they plan to watch a movie later. However, we find many such numbers decrease on the release date, suggesting that a lot of users who had added the movie to the list changed their status to having watched the movie. Using this information, we compute the watchlist size change on the

online release date for each movie and compare the average values between two sets of movies: those with a release gap of less than 30 days and those with a release gap of more than 30 days. We exclude movies having overlapped release windows between the two stages to mitigate the impact of cross-stage releases. Table 8 presents the results.

We separate the comparisons between Chinese and American movies. The drop in the watchlist size for movies released within 30 days after their theatrical run is much larger (i.e., − 54.30 for Chinese movies and − 34.75 for American movies) than that for movies with a longer release gap (i.e., − 8.86 for Chinese movies and − 9.90 for American movies), which indicates that the behavioral change is higher when the release gap is smaller. The one-tailed t-tests suggest that the difference between the two comparisons is significant (p-value < 0.1 for both Chinese and American movies). In other words, more consumers adding a movie to the watchlist during the cinema release stage would convert to watching the movie if it is released shortly after its theatrical run. By contrast, fewer consumers change their behavior when the release gap is substantially large, indicating that many earlier consumers lose interest in the product. These results are consistent with our conjectures. After all, the evidence supports the second explanation.

Given the above findings, we posit that there are potential qualitative differences between the consumer groups at the in-theater stage and the online stage for American movies. In particular, the self-reported viewership at the online stage is more likely to include Douban users living outside of China (e.g., students studying in the US). These users live in an environment that makes them more familiar with Western culture and thus are more interested in American movies. By watching these movies, they can discuss them with social connections in their network. For various reasons, they may not be able to watch the movie in the local market (e.g., high ticket prices in the foreign market). Once the movie is released online (with a low subscription cost), they will watch it and report their consumption experience on Douban. Like the early adopters at the in-theater stage, these individuals have a strong preference for American movies; thus, their decision is affected by review volume only, which is similar to the findings during the early period (i.e., $Age = 1$) of the theatrical run for American movies.

7 Business implications, limitations, and conclusion

This study has investigated the impact of online product reviews on consumer demand over time using movie reviews across two release stages: during the theatrical run and on online streaming platforms. Unlike previous findings of a diminishing sales effect of product reviews (e.g., Liu, 2006), we observe an increasingly positive

Table 8 Changes in watchlist size on the first day of online release

Release Gap	0–30 days	After 30 days
Chinese Movies	− 54.30	–8.86
American Movies	–34.75	–9.90

impact of product reviews in the cinema release stage. Our empirical probe is built upon the view of diffusion of innovations (Rogers, 1962), in which late consumer groups are more conservative and rely heavily on early buyers' opinions.

We also compare the impact of online reviews between two culturally different products – domestically produced Chinese movies and imported American movies – in the Chinese market. The results suggest that consumers are more conservative in selecting foreign products due to their lack of cultural background and knowledge to appreciate them. Late adopters during the cinema release stage only watch American movies with a high review valence *and* volume.

Across the two stages, we find that the inter-release gap may lead to a role change in online reviews. When the release gap is short, product reviews continue their impact on consumer demand during the cinema release stage. By contrast, when the gap is substantially long (e.g., more than 30 days), those planning to watch the movies during the cinema release stage may lose interest in the product due to its unavailability. Consequently, the online release of the product after a long release gap will attract a different consumer group. They tend to select movies with a high review volume (instead of high valence), resulting in a qualitatively different role of product reviews in the online stage.

7.1 Business implications

The empirical findings offer several business insights. First, many sellers focus on soliciting positive reviews during the early period of a product release because they believe this is the most critical period when online WOM affects sales. Our findings discourage such business practices. Only products that receive persistent positive reviews can maintain a high consumer demand. Although soliciting positive reviews during the early period of product release may help improve sales, the product sales will drop quickly if more negative reviews surface in the late period, which may even shorten the product's life cycle. Thus, we suggest sellers focus on improving the product quality so that consumer groups can favor it over time. Second, the results about the consumers' more restrictive selection criteria for products produced in a different culture imply a more competitive environment for foreign products. A cultural product that cannot survive in its home market (e.g., a Hollywood flop) is unlikely to be successful in a foreign market.

Third, we suggest that sellers of products that receive good WOM in an early market shorten the release gap between the two stages. In this way, the products can continuously benefit from the online buzz while consumer interest remains. Conversely, sellers of products with low review valence may consider a longer cross-stage release gap. Such a strategy may help them avoid the negative impact of review valence on consumer demand in the following market. In this case, those with a higher review volume might benefit more from the WOM in the second stage.

7.2 Limitations and future directions

Our study suffers from a few limitations. First, we acknowledge that estimations using user-reported viewership at the online stage may not be accurate if the

outcome does not fully represent the actual demand on the video streaming platforms. Second, although we include many factors, such as Baidu Trends and the number of competing movies, to account for potential factors that may affect both product review measures and the dependent variables, we are unable to address the concern of endogeneity completely. Unlike the setting in Chintagunta et al. (2010), where products are likely to be released sequentially across local markets, all movies in our data are released across the nation on the same date in China. As a result, finding instrumental variables using review measures in the previous market is not an option. We are cautious that the empirical findings in this study do not suggest causal effects.

Lastly, we are unable to measure repeat consumption in our study. A true lover of a movie may watch it several times. Our estimations of the impact of product reviews on consumer demand do not account for this scenario. We encourage future studies to address the related questions. Despite these limitations, we believe this study sheds light on the temporal role of product reviews from novel perspectives, i.e., across markets and over time.

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Declarations

Ethics approval Not applicable.

Informed consent Not applicable.

Consent for publication Yes.

Competing interests The authors declare no competing interests.

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